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| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 04/17/2008 |
| **PI Name:** | Roux, Stanley Ph.D. | | |
| **Project Title:** | Cellular bases of light regulated gravity responses | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 78712-1240 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology<br>01-OBPR-06 |
| **Start Date:** | 11/01/2002 | **End Date:** | 10/31/2007 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 9 |
| **No. of Bachelor's Candidates:** | 9 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE through 10/31/2007, per A. Chu (9/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Clark, Gregory ( The University of Texas at Austin ) | | |
| **Grant/Contract No.:** | NAG2-1586 | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | The overall goal of this project is to clarify the cellular mechanisms that control light-regulated gravity responses. Our main aim this year was to advance the overall objective of this project by providing new information on steps in the transduction chain for gravity-regulated responses in which Ca2+ is likely to exert a critical influence. Because earlier discoveries revealed that plant cells release ATP into their walls as they grow, that this extracellular ATP (eATP) can induce calcium signaling changes in plants and regulate gravitropic growth, and that ectoapyrase enzymes that control the accumulation of extracellular nucleotides play a major regulatory role in growth control, our specific objective this year was to elucidate the calcium-dependent signal transduction pathway connecting extracellular nucleotides to growth changes oxidase. A second objective was to determine whether blocking the expression of ectoapyrase enzymes significantly reduces the downward curvature of roots. Finally a third objective was to develop a method for measuring the [eATP] in plant cell walls to determine whether the differential growth that occurs on the two sides of root cells during root gravitropism results in a differential accumulation of eATP in the walls of the faster growing (top side) and slower growing (bottom side) cells of the root as it curves downward. Findings from these experiments would provide valuable new insights into another (and unexpected) regulatory agent that controls gravitropic growth. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research does not directly seek to understand a disease or malady that affects humans on Earth and/or in space, nor does it seek to develop new therapeutics for alleviating symptoms of a malady on Earth. However, the findings will help to understand the growth of plants better, and since plants are a crucial source of food for humans, this research does seek to understand the malady of malnutrition. Also, this research does yield a new understanding of basic biological processes, specifically the processes of plant growth and the cellular mechanisms whereby gravity can affect the developmental polarity in cells. Further, this research points to a real role of gravity in regulating growth and development of plants on Earth and thus reveals potential problems in achieving normal growth and development of plants in space. Finally, the health of humans is inextricably linked to the ability to control and continuously improve the growth of plants. This, in turn, requires an improved understanding of the molecular mechanisms that control growth in plants. The accomplishments of this research contribute to that improved understanding, and thus indirectly benefit humans. |
| **Task Progress:** | During this past, and last, year of the project the main area of progress employed the model flowering plant Arabidopsis, In the Arabidopsis studies, our earlier work showed that treatment of Arabidopsis seedlings with ATP inhibited the gravitropism of their roots at concentrations that did not inhibit root growth. Subsequently we focused on the question of what signaling steps converted the eATP stimulus into a growth response.<br>During this past year we found that sub-micromolar concentrations of ATP could induce the production of superoxide, and that this effect could be blocked by agents that reduced the uptake of calcium into cells. Superoxide is rapidly converted to hydrogen peroxide, a growth regulator recently implicated in gravitropism control. Ethylene production is typically downstream of the superoxide/hydrogen peroxide signal, so it is possible that the effects of eATP on ethylene production are mediated by its earlier induction of superoxide, which, in turn, is dependent on eATP induction of increased uptake of calcium into cells, presumably through ATP-activated calcium channels. Subsequent reports by others have verified that superoxide and ethylene are intermediate signaling agents important for the control of gravitropic growth.<br><br>The above data suggest that accumulation of eATP in the plant cell wall above a certain threshold would inhibit growth. Growth is typically accompanied by the delivery of secretory vesicles to the plasma membrane and by the delivery of their contents into the wall. In animals secretory vesicles are known to contain high concentrations of ATP, and this is likely also in plants. If so, then growth would be normally accompanied by an increase of [ATP] in the walls of cells. If increased [ATP] in the wall inhibits growth, then for growth to be maintained, it would be important to hydrolyze the wall ATP accumulating in growth zones. Consistent with this idea, the level of mRNA encoding the eATP-hydrolyzing ectoapyrase enzyme is highest in tissues with the highest growth rates. Stimuli that result in growth reduction (e.g., red light irradiation of etiolated seedlings) result simultaneously in reduction of apyrase mRNA levels.<br><br>Taken together the results described in this section lead to the novel insight that growth control in plants involves, in part at least, control of eATP accumulation by the activity of ectoapyrases. They predict that genetic suppression of apyrases could suppress the gravitropic growth response. The function of two Arabidopsis apyrases, AtAPY1 and AtAPY2, has been shown to participate importantly in the control of growth. These two apyrases are 87% identical in protein sequence, have signal peptides, and closely resemble a pea ectoapyrase. They also co-purify with plasma membranes in cell fractionation assays, and most of their structure appears to be accessible to protease digestion when purified plasma membranes that are in an outside-out orientation are treated with trypsin. A role for these enzymes in growth control was demonstrated by Wu et al. (Plant Physiology 2007), who showed that mutants knocked out in both apyrases cannot grow.<br><br>Both AtAPY1 and AtAPY2 are expressed in roots, and we found that 3-d-old light-grown seedlings of both Atapy1 and Atapy2 mutants show a significantly reduced rate of downward curvature of their primary roots when gravistimulated by being rotated to a horizontal position. For these experiments curvature measurements were taken 24 h after gravistimulation. The suppressed gravitropism of the mutants did not appear to be due to reduced growth of the primary root, for this growth was the same in 3-d-old mutants and wild-type plants. If we assume that AtAPY1 and AtAPY2 function as ectoapyrases to reduce [eATP], these results are consistent with an earlier finding that increased eATP (applied to the growth medium) inhibits root gravitropism. The mechanism by which accumulations of eATP inhibit gravitropism is not known, but may be related to the ability of high [eATP] to inhibit the transport auxin, a key hormone that regulates gravitropism, as reported earlier by Tang et al.<br><br>A question that arises from the above studies is how much ATP is actually present in the extracellular matrix (= cell wall) of plant cells. To address this question we have generated transgenic plants expressing a wall-directed luciferase enzyme. Luciferase is a firefly enzyme that, in the presence of luciferin, gives off light proportional to the [ATP] in its environment. Thus wall luciferase can serve as a reporter molecule to measure the [ATP] in the wall of the plant cells. Using plants expressing wall luciferase we have verified that rapidly growing tissue has more [eATP] than slower growing tissue, and that, specifically the [eATP] in the wall of cells on the upper side of a gravitropically bending root is higher than that in the wall of cells on the slower growing lower side. We now aim to calibrate the light given off by ATP-reacting wall luciferase to quantify what is the [eATP] in the ECM of plant cells. This tool will allow us to |

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| | quantitatively evaluate the role of eATP in regulating gravitropic growth. |
| **Bibliography Type:** | Description: (Last Updated: 10/16/2023) |
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